SHIP RESISTANCE · METHOD COMPARISON

Holtrop–Mennen vs CFD

Two independent predictions of the calm-water resistance of the Variant 22 bulk carrier, compared speed by speed

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Holtrop–Mennen predicts about twice the CFD resistance at 2.3 and 4.6 m/s.

1/8
The CFD skin friction is about one eighth of the standard ITTC-1957 value.

2–11 %
Remaining gap at low speed once the CFD friction is brought up to the standard value.

Fn 0.3+
Above this speed the methods diverge sharply, and neither is reliable at Fn 0.4–0.5.

Summary

At the speeds a ship of this size would normally sail, the two methods tell a consistent story once one known weakness of the CFD model is accounted for.

Holtrop–Mennen predicts a bare-hull resistance of 7.4 kN at 2.3 m/s and 31.3 kN at 4.6 m/s, roughly twice the CFD values of 3.7 and 14.7 kN. Almost all of that difference is skin friction: the CFD friction force is consistently about one eighth of the long-established ITTC-1957 value. When the CFD friction is replaced with the standard value, the two methods agree to within 11 % at 2.3 m/s and 2 % at 4.6 m/s.

At 6.9 m/s and above, the methods separate quickly. The CFD resistance coefficient stays constant, which means the simulation is not picking up the wave system that a full hull creates at these speeds. Holtrop–Mennen, on the other hand, is being used beyond the speeds it was built from. Results at Fn 0.4 and 0.5 should therefore be read as trends rather than reliable figures, whichever method they come from.

1 Purpose of this comparison

Our CFD study of the Variant 22 hull ended with a clear recommendation: check the results against an established empirical method before relying on them. This document does exactly that. It sets the ANSYS Fluent results beside a Holtrop–Mennen estimate for the same hull, at the same five speeds, and explains where the two agree, where they do not, and why.

Holtrop–Mennen was chosen because it is the method most widely used in early-stage ship design. It needs only the main dimensions and a handful of hull-form coefficients, all of which the offset table already gives us.

2 The ship

Variant 22 is a small, full-bodied bulk carrier with a long parallel middle body, a flat bottom and blunt ends. It has no bulbous bow. All particulars below come from the offset table and the CFD report.

Particular	Value	Particular	Value
Length between perpendiculars	54.0 m	Block coefficient	0.759
Breadth	11.0 m	Prismatic coefficient	0.777
Draught (even keel)	5.0 m	Midship coefficient	0.977
Displaced volume	2,254 m³	Waterplane coefficient	0.903
Wetted surface	946 m²	Centre of buoyancy	1.9 m aft of midships

Table 1. Principal particulars of Variant 22, taken from the offset table and the CFD report.

The hull is a good fit for Holtrop–Mennen in terms of fullness: its prismatic coefficient of 0.777 lies inside the range the method was derived from for bulk carriers and tankers. It is a little shorter and a little shallower for its breadth than the ships in that data set (length-to-breadth 4.9 and breadth-to-draught 2.2, against typical minimums of 5.1 and 2.4), so the estimate carries its usual uncertainty of around ±10 %.

3 The two methods in brief

CFD: ANSYS Fluent

Solves the flow of water and air around the hull directly. The free surface is tracked, so waves can form on their own.

Bare, smooth hull in fresh water; k–ω turbulence model; 500 iterations at each speed.

Reports a pressure force and a friction force.

Empirical: Holtrop–Mennen (1982)

Statistical formulas fitted to hundreds of model tests and sea trials at MARIN, the Netherlands ship model basin.

Builds the total from separate parts: friction, the extra drag caused by hull shape, wave making, appendages, bulb, transom and an allowance for roughness and air.

Gives full-scale values from main dimensions alone.

Making it a fair comparison. Both methods use the same hull data, fresh water at the CFD's density and viscosity, the same wetted area of 946 m², and no rudder or other appendages. Holtrop–Mennen's allowance for roughness and air is shown separately, because a smooth-hull CFD model has nothing equivalent. The estimates were produced with our Holtrop–Mennen calculator, which reproduces the published worked example of the method exactly.

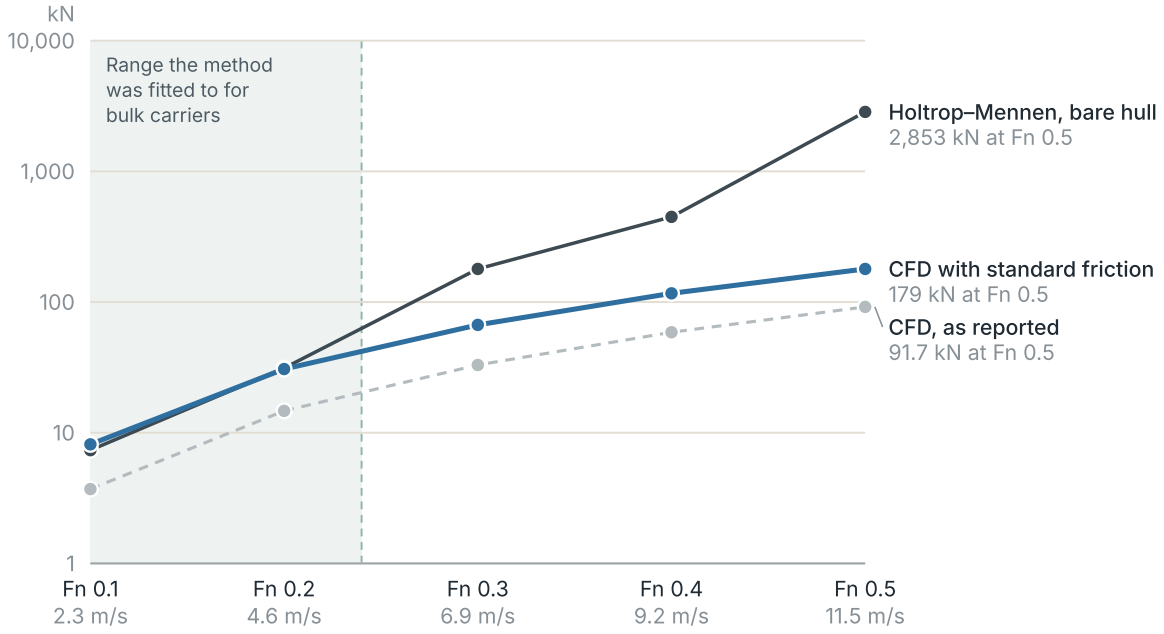
4 Results side by side

At every speed Holtrop–Mennen predicts more resistance than the CFD. At the two lower speeds the difference is about a factor of two. Above Fn 0.3 the gap widens rapidly.

Speed (m/s)	Speed (knots)	Froude number	CFD (kN)	Holtrop–Mennen, bare hull (kN)	Holtrop–Mennen, with allowance (kN)	Ratio, bare hull
2.3	4.5	0.1	3.7	7.3	8.9	2.0×
4.6	8.9	0.2	14.7	31.3	37.6	2.1×
6.9	13.4	0.3	33.0	178.9	193.1	5.4×
9.2	17.9	0.4	58.7	448.7	473.9	7.6×
11.5	22.4	0.5	91.7	2,852.9	2,892.2	31.1×

Table 2. Total resistance at the five CFD speeds. CFD values from Table 5 of the CFD report. The ratio compares like with like: bare hull against bare hull.

Figure 1. With standard friction, the CFD matches Holtrop–Mennen up to Fn 0.2, then falls well short



Bare-hull resistance in kN on a logarithmic scale. "CFD with standard friction" keeps the CFD pressure force and replaces its friction force with the ITTC-1957 value.

5 Where the difference comes from

5.1 Skin friction is the main gap at low speed

The friction force in the CFD is between 12 % and 14 % of the standard ITTC-1957 friction line at every speed. The CFD report already identified the likely reason: the mesh next to the hull has no inflation layer, so the thin boundary layer where friction is generated is not resolved. This one effect accounts for nearly the whole factor of two at 2.3 and 4.6 m/s.

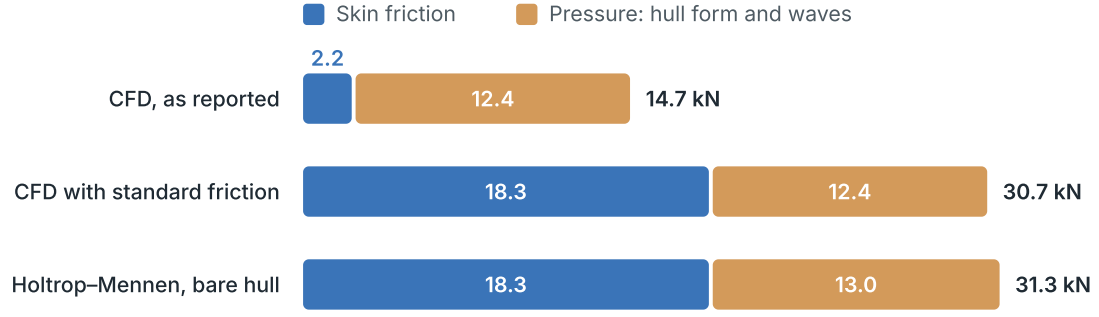
Froude number	CFD friction (kN)	Standard friction, ITTC-1957 (kN)	CFD as a share of standard
0.1	0.6	5.05	12 %
0.2	2.2	18.33	12 %
0.3	5.1	39.06	13 %
0.4	9.0	66.88	13 %
0.5	14.0	101.54	14 %

Table 3. Friction force from the CFD compared with the standard ITTC-1957 friction line for the same wetted area.

5.2 The pressure force agrees well

The more encouraging finding is that the CFD pressure force, which captures the drag caused by the hull's shape and by the waves it makes, matches Holtrop–Mennen closely at low speed: 12.4 kN against 13.0 kN at 4.6 m/s, and 3.1 kN against 2.3 kN at 2.3 m/s. The large pressure share reported by the CFD is therefore largely genuine. It reflects a blunt, full hull with a short after body, not just a numerical artefact.

Figure 2. At 4.6 m/s the methods differ in friction, not in pressure



Resistance in kN at 4.6 m/s (8.9 knots, Fn 0.2). The Holtrop–Mennen allowance for roughness and air (6.3 kN) is not included.

5.3 Wave making is missing from the CFD at higher speeds

The CFD resistance coefficient stays at about 1.47×10^{-3} at all five speeds, so its resistance simply grows with the square of speed. A full hull should behave differently: wave making rises steeply beyond about Fn 0.25. At 6.9 m/s Holtrop–Mennen attributes 122 kN to waves alone, more than four times the CFD's entire pressure force. The likely causes, both noted in the CFD report, are a free-surface mesh too coarse to hold the waves and a steady-state solver.

5.4 Holtrop–Mennen is outside its range at the top speeds

For bulk carriers, the formulas were derived from ships tested up to about Fn 0.24. The Fn 0.4 and 0.5 cases are far beyond that and beyond any realistic service speed for this hull. Holtrop–Mennen's 2,853 kN at Fn 0.5 correctly signals that resistance would be very high, but it should not be quoted as a figure.

6 Effective power

Effective power follows the same pattern. At around 9 knots (4.6 m/s), a realistic speed for a ship of this size, the corrected CFD and Holtrop–Mennen both point to roughly 140–175 kW, compared with 67 kW from the CFD as originally reported.

Speed (m/s)	Speed (knots)	CFD, as reported (kW)	CFD with standard friction (kW)	Holtrop–Mennen (kW)
2.3	4.5	8	19	20
4.6	8.9	67	141	173
6.9	13.4	228	462	1,332
9.2	17.9	540	1,073	4,360
11.5	22.4	1,054	2,060	33,261

Table 4. Effective power, P_E = resistance × speed. The Holtrop–Mennen column includes the allowance for roughness and air, as the method defines it. Values at Fn 0.4 and 0.5 are indicative only.

7 How sure are these numbers?

We tested how much the Holtrop–Mennen estimate moves when its assumptions change. Most choices shift the result by a few percent. One does not: whether the stern of the CAD model leaves a flat transom below the waterline.

What changes	Resistance at 4.6 m/s (kN)	Effect
Baseline assumptions (Section 2)	37.6	—
Hull ends squarely at AP, leaving about 24 m² of transom below the waterline	76.2	+103 %
Standard formulas for wetted area and bow angle instead of values from the offsets	39.9	+6 %
Sea water instead of fresh water	39.2	+4 %
Fuller, U-shaped stern sections	38.4	+2 %
Finer, V-shaped stern sections	36.8	−2 %

Table 5. Sensitivity of the Holtrop–Mennen total at 4.6 m/s (Fn 0.2).

> **Worth checking first.** The offsets show the hull is still 8 to 10.5 m wide at the aft perpendicular, between the 3 m and 5 m waterlines. If the CAD model simply ends there, it has a flat transom of about 24 m² under water. That would add around 11 kN of drag at 2.3 m/s, three times the whole CFD result, and would double the Holtrop–Mennen estimate at 4.6 m/s.

On the CFD side, the report notes that two different solver settings gave answers 9 % apart, and that a single mesh was used. CFD figures are therefore best read to two significant figures.

8 Conclusions and recommendations

- **The two methods agree where it matters most.** At 2.3 and 4.6 m/s, once the CFD friction is corrected, they are within 11 % and 2 % of each other.
- **The CFD captures hull-form drag well** but underestimates skin friction by a factor of about eight.
- **The CFD does not yet capture wave making**, so its results above Fn 0.25 are too low.
- **A realistic bare-hull resistance at 4.6 m/s is about 31 kN**, or about 38 kN including the usual allowance for roughness and air.

Recommended next steps, in order of priority:

1. Confirm how the stern is closed in the CAD model at the aft perpendicular.
2. Add an inflation layer along the hull and rerun 2.3 and 4.6 m/s. The friction force should rise to about 5 and 18 kN.
3. Rerun 6.9 m/s as a transient case with a finer mesh along the free surface. Holtrop–Mennen expects about 120 kN of wave resistance there.
4. Check the reference values set in Fluent, as recommended in the CFD report.
5. Treat 9.2 and 11.5 m/s as qualitative cases in both methods.

A Appendix: Holtrop–Mennen details

The estimate follows the eight-step procedure of Holtrop and Mennen (1982). The terms that matter most for this hull:

Quantity	Value	Comment
Length of run	7.85 m	Short: the after body tapers quickly
Form factor, $1 + k_1$	1.456	Upper end of the usual range for full hulls
Half angle of entrance	22.8°	Measured from the offsets at the design waterline
Bulb and transom factors	1.0 and 1.0	No bulb; stern assumed clear of the water
Correlation allowance, C_A	6.30×10^{-4}	Standard hull roughness of 150 µm

Table A1. Key Holtrop–Mennen terms for Variant 22. They do not depend on speed.

Speed (m/s)	Froude number	Friction (kN)	Friction with form (kN)	Waves (kN)	Allowance (kN)	Total (kN)
2.3	0.1	5.05	7.35	0.00	1.57	8.92
4.6	0.2	18.33	26.69	4.60	6.29	37.59
6.9	0.3	39.06	56.88	122.07	14.16	193.11
9.2	0.4	66.88	97.38	351.30	25.18	473.86
11.5	0.5	101.54	147.85	2,705.05	39.34	2,892.24

Table A2. Holtrop–Mennen components at the five CFD speeds. Appendage, bulb and transom terms are zero for this hull. Fresh water, $\rho = 998.2$ kg/m³, $\nu = 1.004 \times 10^{-6}$ m²/s.

References

1. Singhal, A., Vyas, R. and Kakar, R. (2026). *Resistance of a Bulk Carrier Hull Across Five Froude Numbers (Variant 22)*. IIT Kharagpur.
2. Holtrop, J. and Mennen, G. G. J. (1982). An approximate power prediction method. *International Shipbuilding Progress*, 29(335), 166–170.
3. ITTC (1957). Model–ship correlation line. Proceedings of the 8th International Towing Tank Conference.
4. Holtrop–Mennen resistance calculator, github.com/Playtozon/holtrop-mennen-calculator.